# Statistical Case Studies, Project 1 Part 1, Code

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## Executive Summary

## Data

- Description of data

### Filtering and Cleaning

Table 1: Table 1 — quality codes and their frequency in the data

| Code number | Code name | Abbreviated code name | Count | Percentage |
|---|---|---|---|---|
| 50 | Good | G | 336,175 | 95.89% |
| 100 | Estimated | E | 14,388 | 4.10% |
| 140 | Provisional | Prov | 0 | 0.00% |
| 150 | Suspect | S | 0 | 0.00% |
| 200 | Unchecked (imported from legacy database) | V | 0 | 0.00% |
| 254 | Unchecked | U | 30 | 0.01% |

- Justify why kept 100 and above but not 254

### Exploratory Analysis

Throughout the report, we add three time variables to support our analysis: the month, the season, and the water year. The water year runs from September to August and is labelled by the year it ends in, which keeps each winter's high flows together in the same year.

#### Main time series and summary table

We first construct the hydrograph for Don at Parkhill over our chosen period.

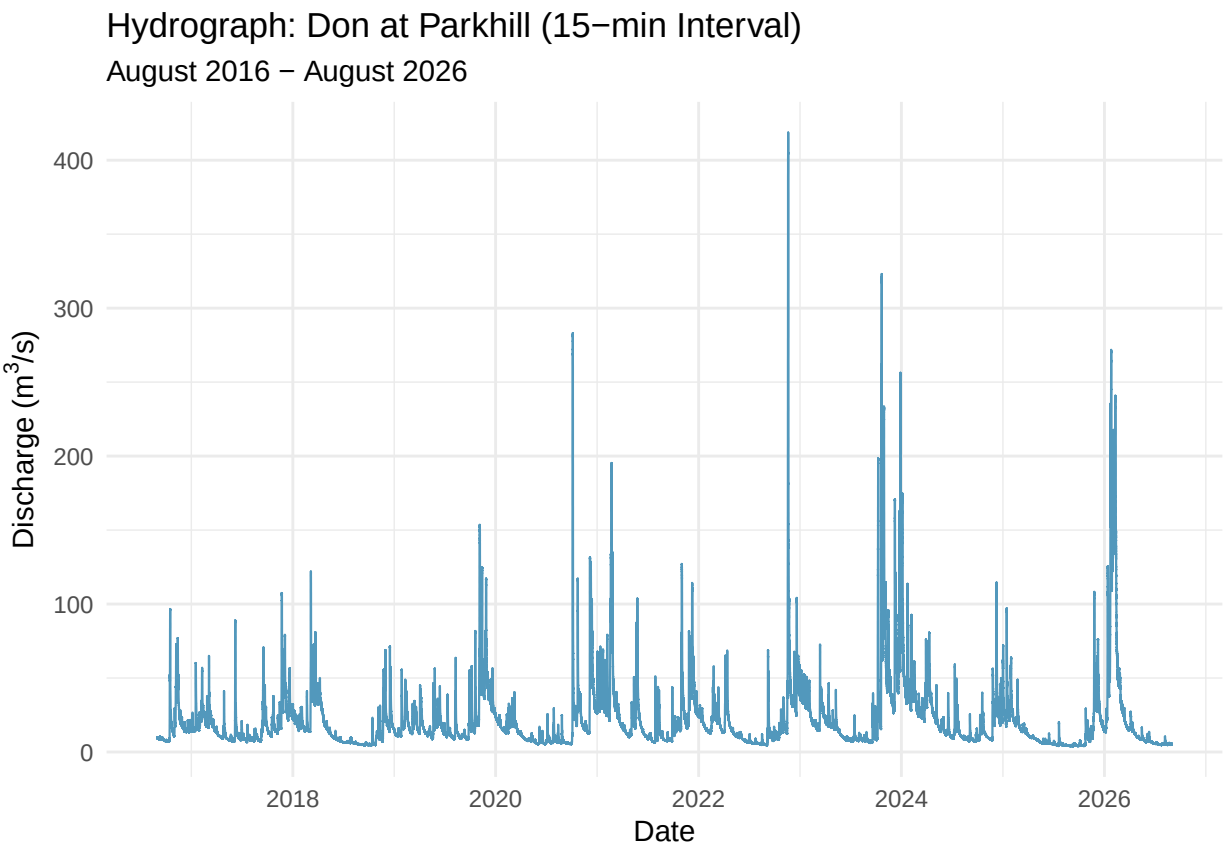
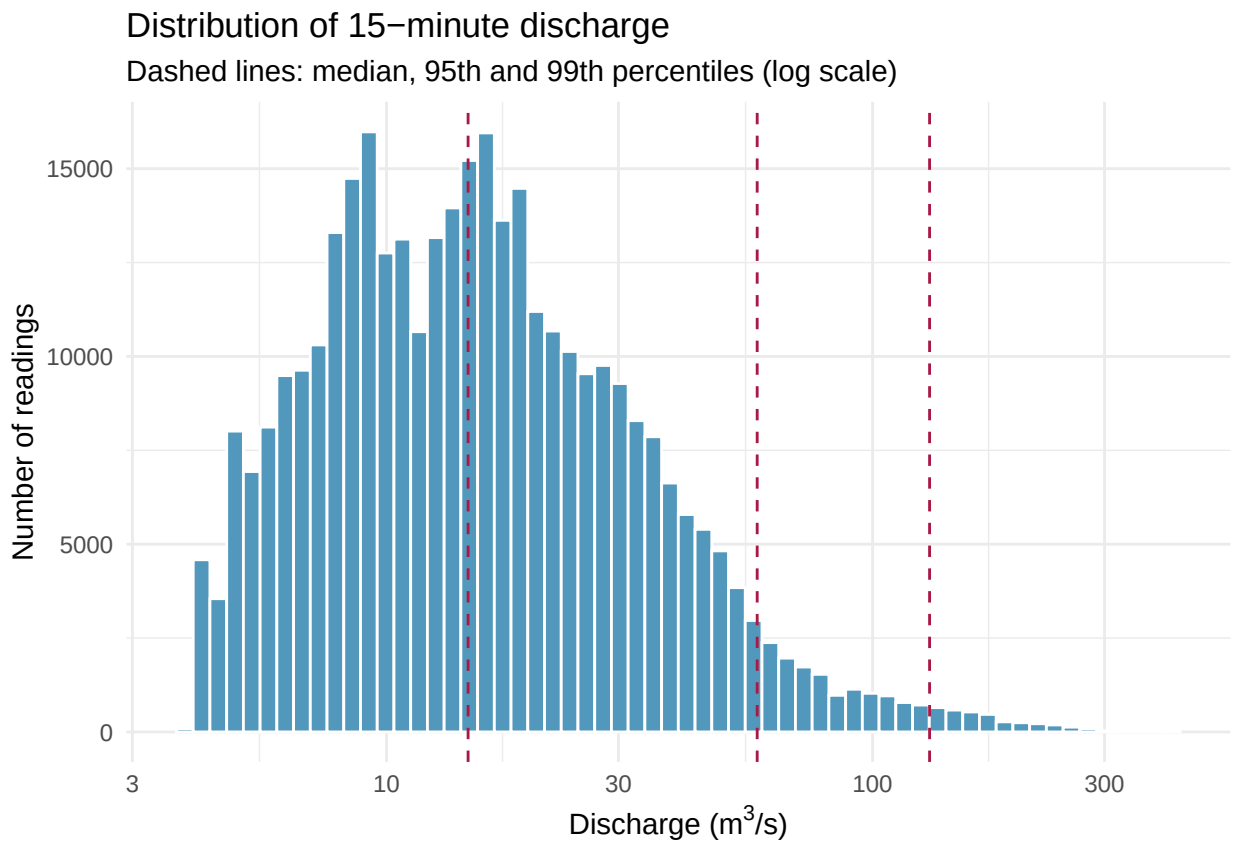



Table 2: Summary statistics for 15-minute discharge ($m^3/s$)

| Statistic | Value |
|---|---|
| Mean | 21.67 |
| Standard deviation | 24.37 |
| Minimum | 3.90 |
| 25th percentile | 8.79 |
| Median | 14.71 |
| 75th percentile | 25.44 |
| 95th percentile | 57.88 |
| 99th percentile | 130.96 |
| Maximum | 418.68 |



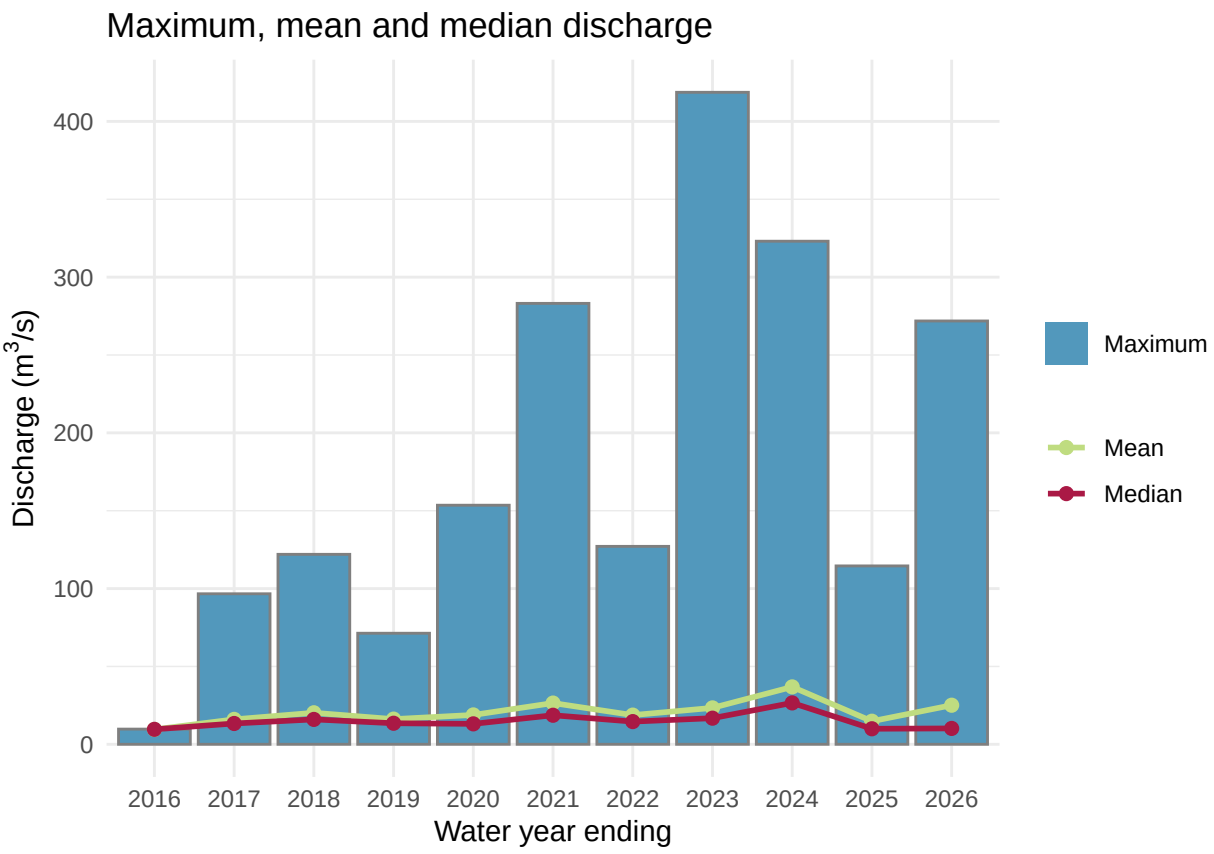

We observe that discharge is low for most of the year, with a smaller number of large peaks. The strong-right skew of the distribution confirms this, with a median discharge of 14.91 $m^3/s$, and a 99th percentile of 130.96 $m^3/s$.

### Yearly trends

From figure ..., we also observe an upward trend in discharge, and larger maximum flows, over the time period.

Table 3: Discharge by water year ($m^3/s$)

| Water year | Mean | Median | Maximum | Date of maximum |
|---|---|---|---|---|
| 2016 | 9.54 | 9.61 | 9.74 | 31 Aug 2016 |
| 2017 | 16.01 | 13.37 | 96.65 | 16 Oct 2016 |
| 2018 | 20.24 | 15.98 | 121.98 | 07 Mar 2018 |
| 2019 | 16.18 | 13.52 | 71.30 | 16 Dec 2018 |
| 2020 | 18.86 | 13.12 | 153.53 | 05 Nov 2019 |
| 2021 | 26.48 | 18.66 | 283.14 | 05 Oct 2020 |
| 2022 | 18.75 | 14.59 | 127.10 | 01 Nov 2021 |
| 2023 | 23.39 | 16.77 | 418.68 | 19 Nov 2022 |
| 2024 | 36.86 | 26.59 | 323.07 | 22 Oct 2023 |
| 2025 | 14.83 | 9.97 | 114.55 | 08 Dec 2024 |
| 2026 | 25.09 | 10.20 | 271.84 | 25 Jan 2026 |


Maximum, mean and median discharge
Discharge (m3/s)
400
300
200
100
0
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Water year ending
Maximum
Mean
Median

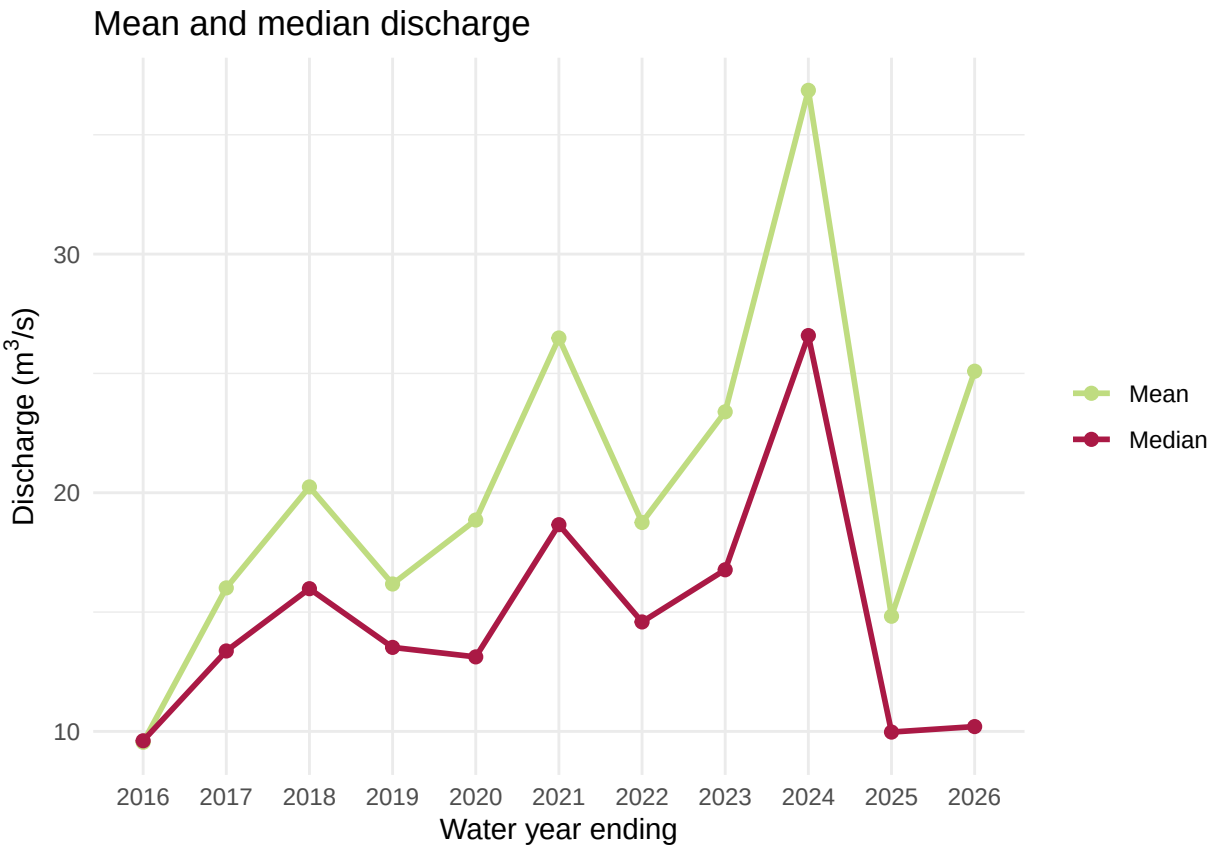



Figure ... provides evidence of this observation, the mean discharge trending upwards from the water year ending in 2017 at ..., peaking in 2024 at .... The maximum discharge follows a similar trend, with both falling sharply in 2025 before rising again.

**Exploring seasonal trends overall**

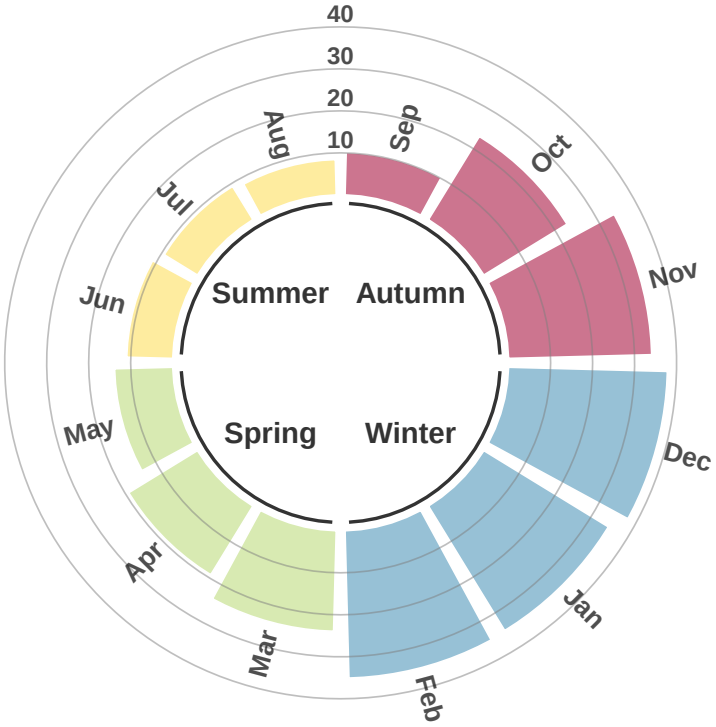
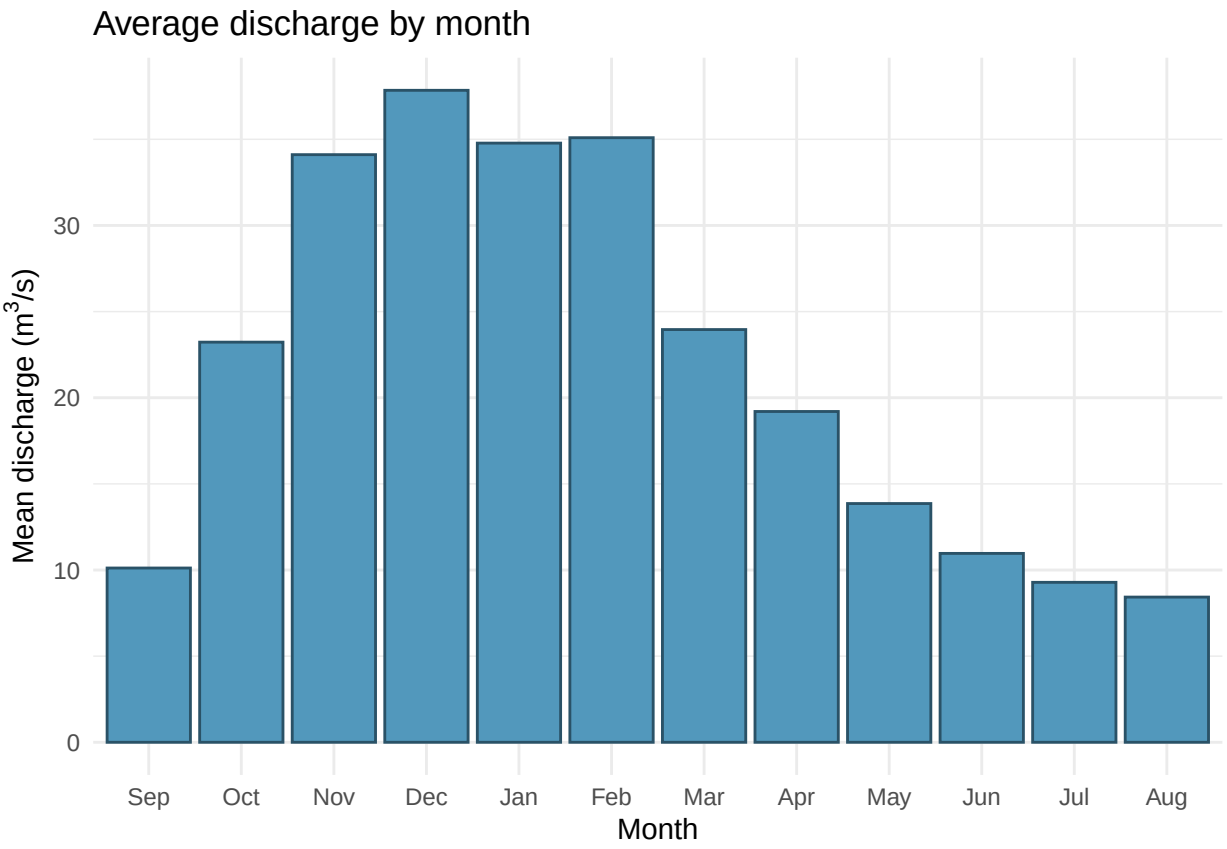
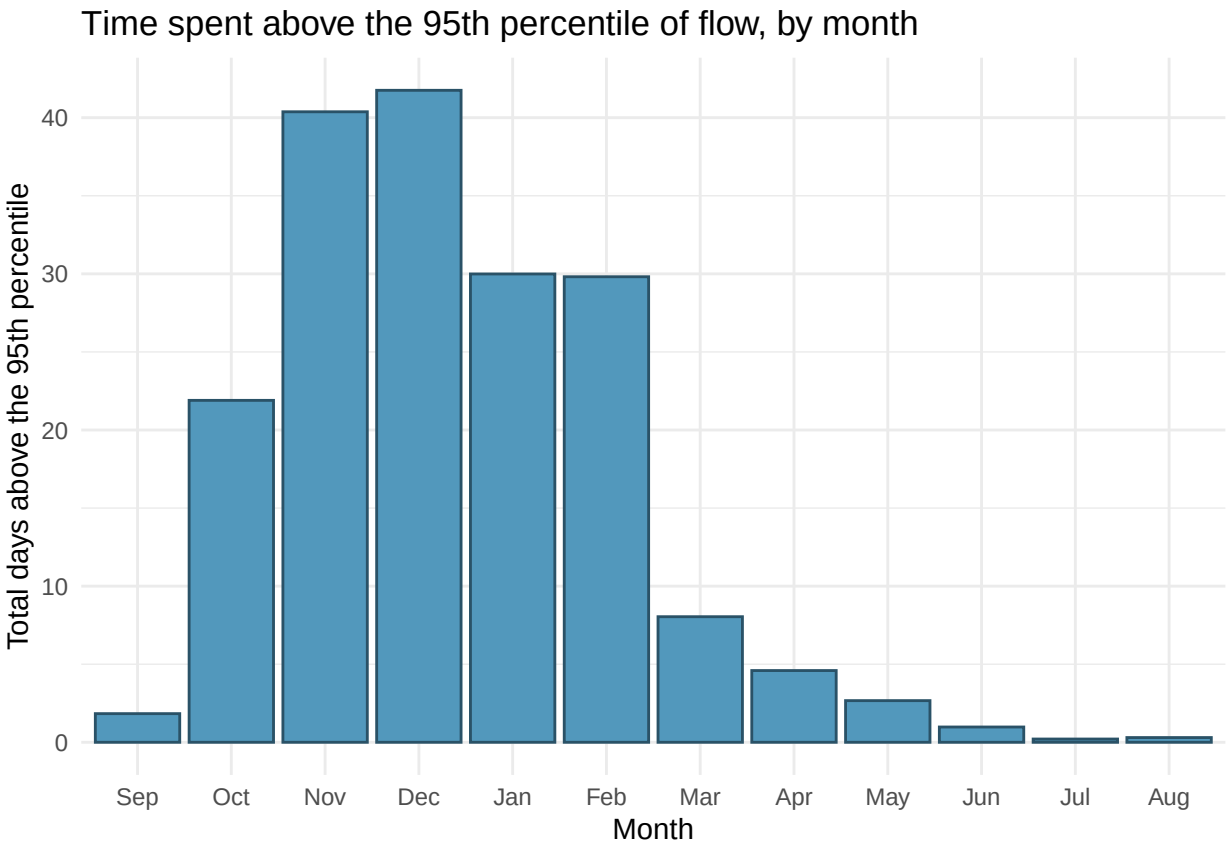
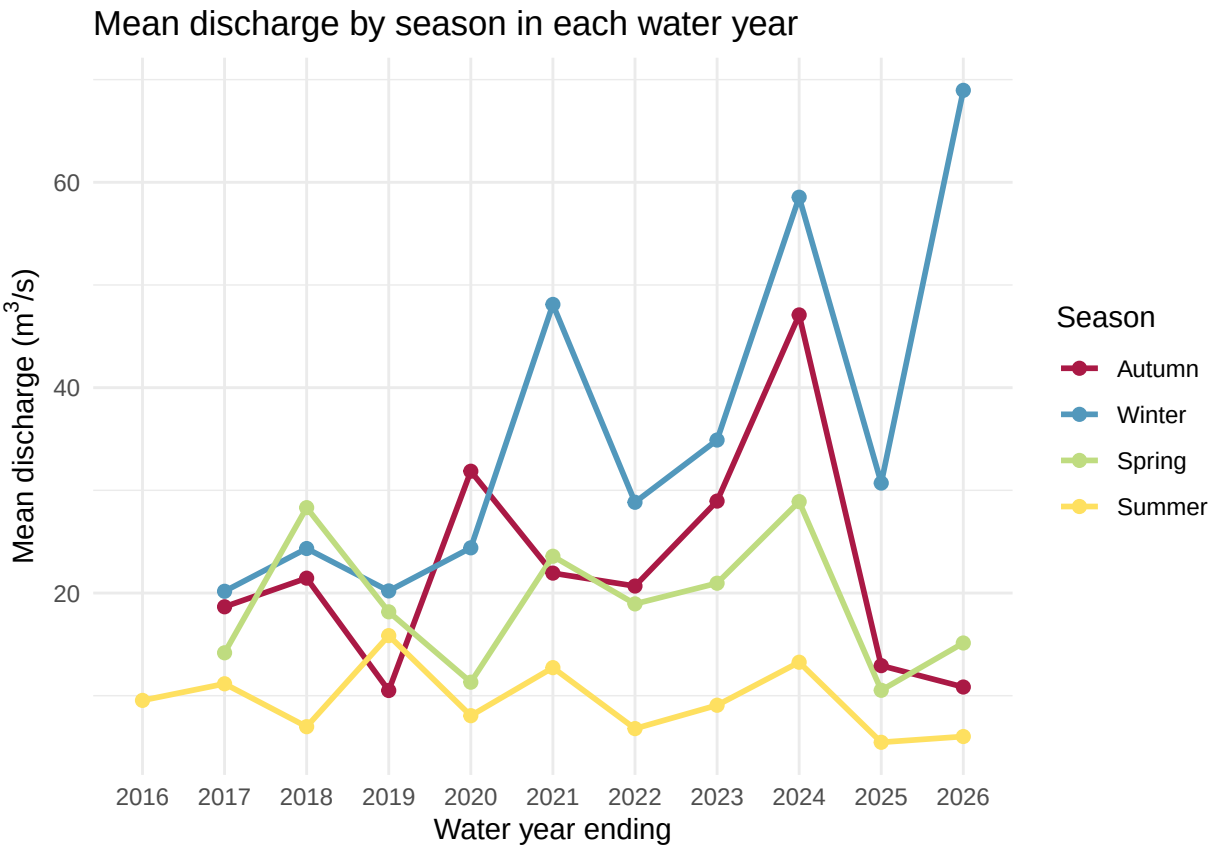


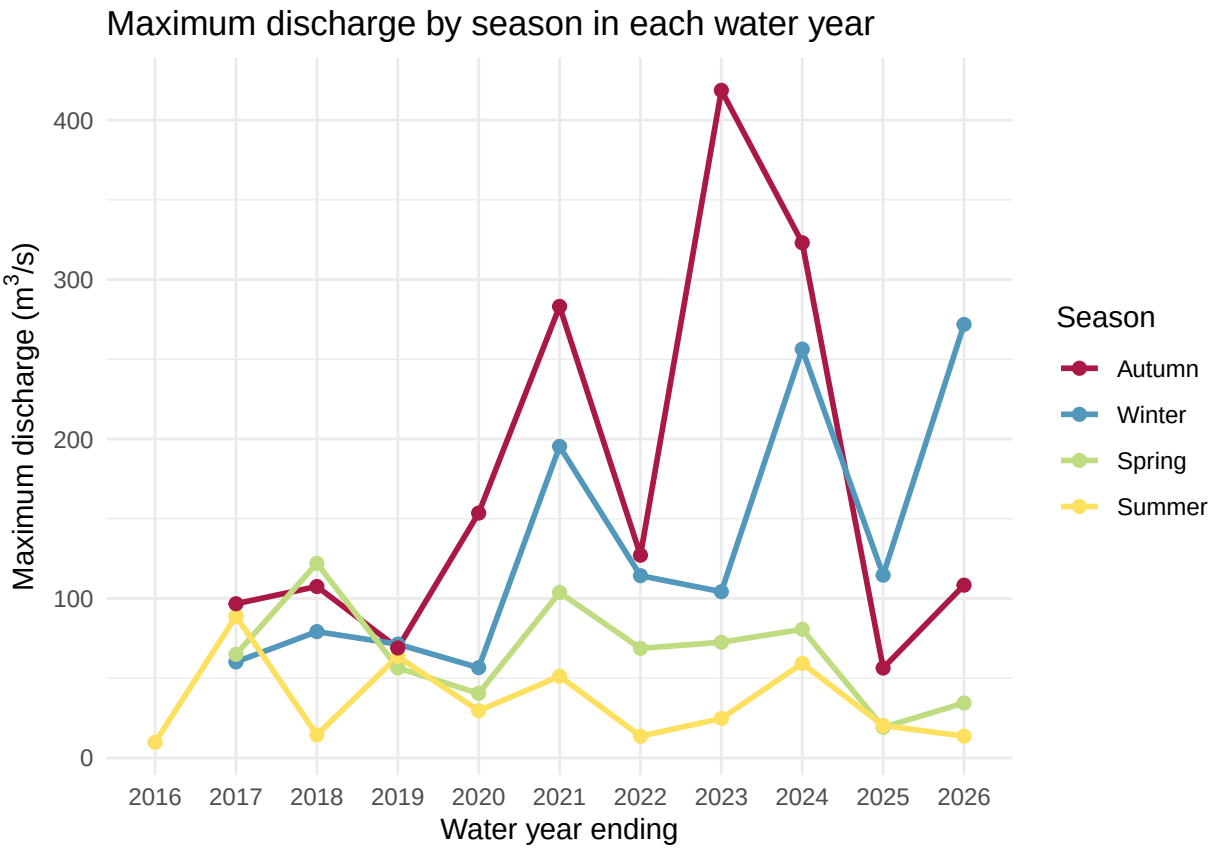

Table 4: Discharge by month ($m^3/s$)

| Month | Mean | Median | 95th percentile | Maximum |
|---|---|---|---|---|
| Sep | 10.12 | 7.91 | 25.34 | 70.53 |
| Oct | 23.22 | 14.49 | 77.46 | 323.07 |
| Nov | 34.11 | 22.40 | 87.85 | 418.68 |
| Dec | 37.84 | 30.60 | 90.53 | 256.33 |
| Jan | 34.78 | 24.78 | 97.58 | 271.84 |
| Feb | 35.09 | 23.90 | 122.03 | 240.84 |
| Mar | 23.95 | 20.55 | 44.76 | 121.98 |
| Apr | 19.20 | 15.31 | 41.15 | 80.63 |
| May | 13.86 | 11.14 | 29.56 | 103.72 |
| Jun | 10.96 | 8.85 | 22.97 | 88.89 |
| Jul | 9.28 | 7.75 | 18.49 | 59.17 |
| Aug | 8.43 | 7.79 | 14.83 | 63.62 |

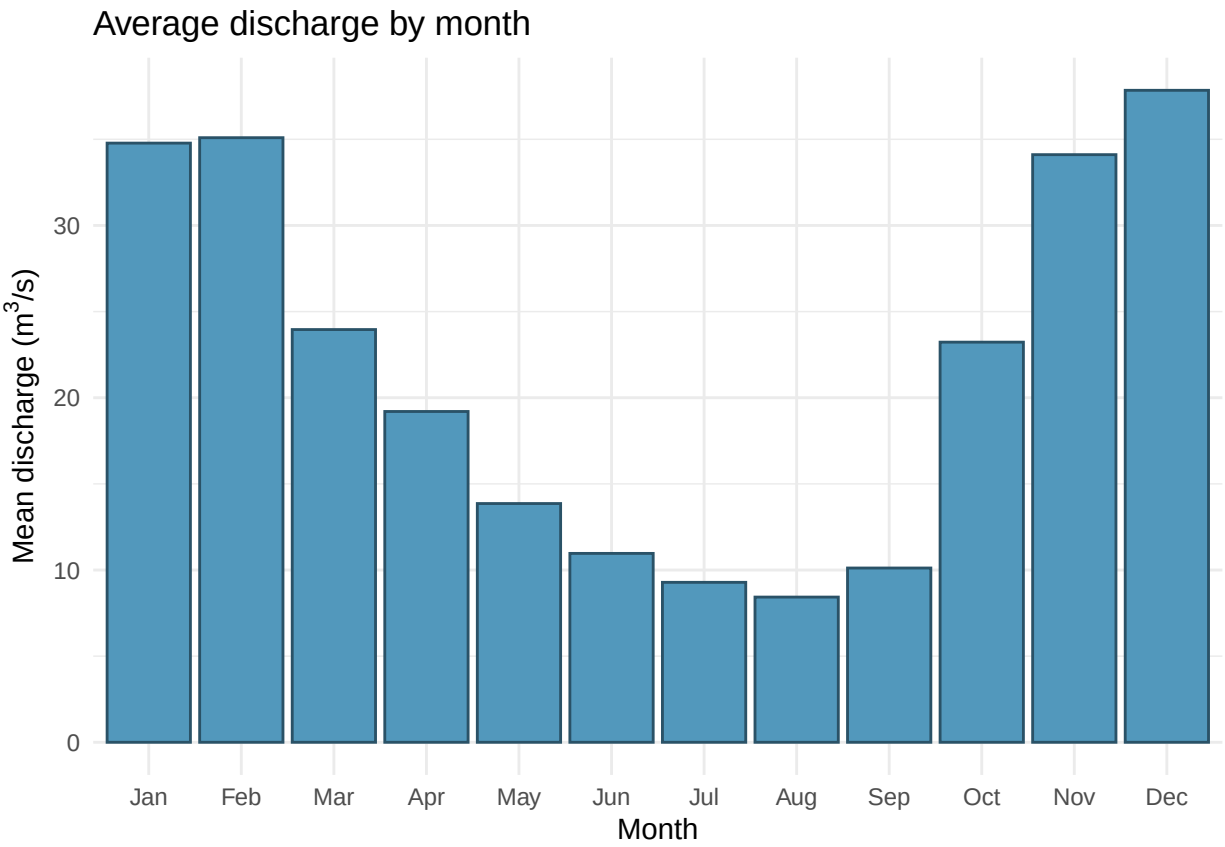

Average discharge by month
Mean discharge (m3/s)
0
10
20
30
Sep
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Month


Time spent above the 95th percentile of flow, by month
Total days above the 95th percentile
0
10
20
30
40
Sep
Oct
Nov
Dec
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Month


Mean discharge by season in each water year
Mean discharge (m3/s)
60
40
20
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Water year ending
Season
Autumn
Winter
Spring
Summer




- Other seasonality options.


Average discharge by month
Mean discharge (m3/s)
0
10
20
30
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month

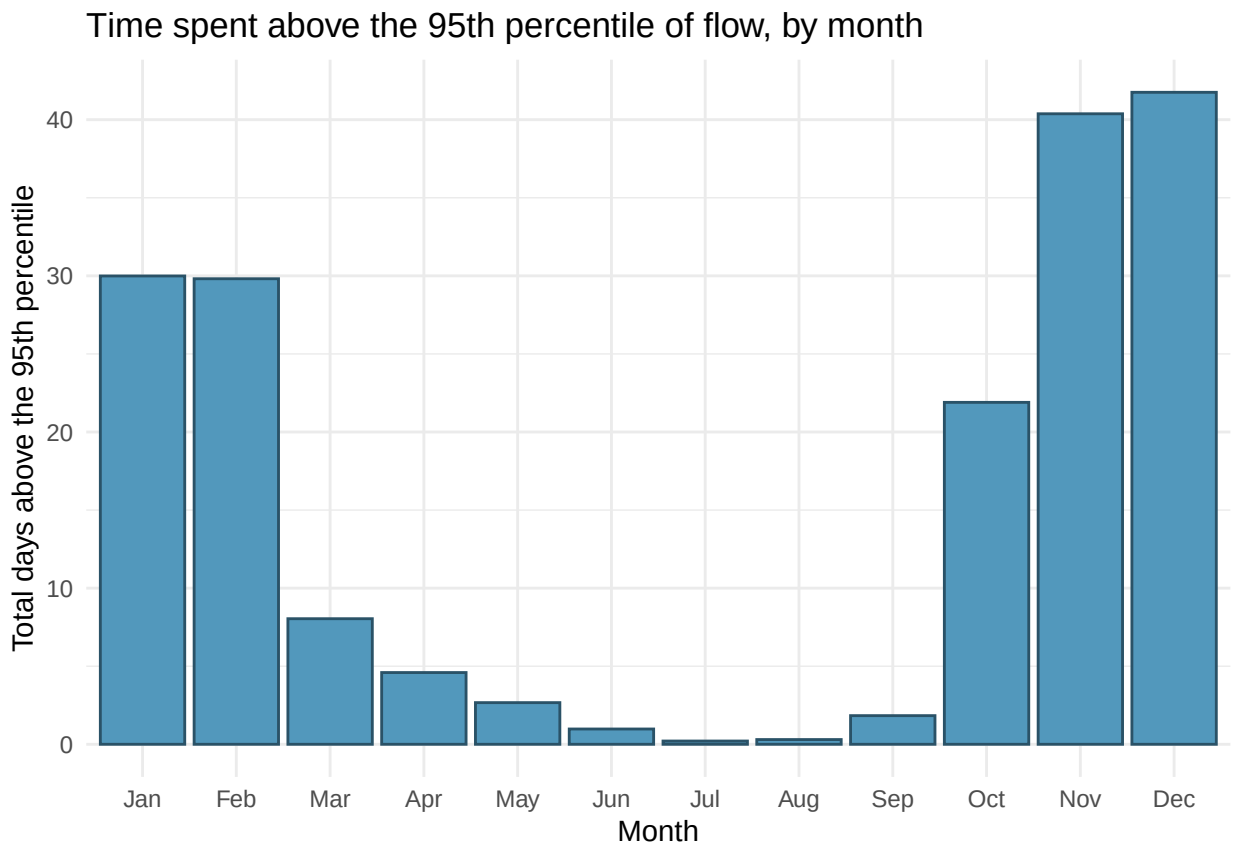
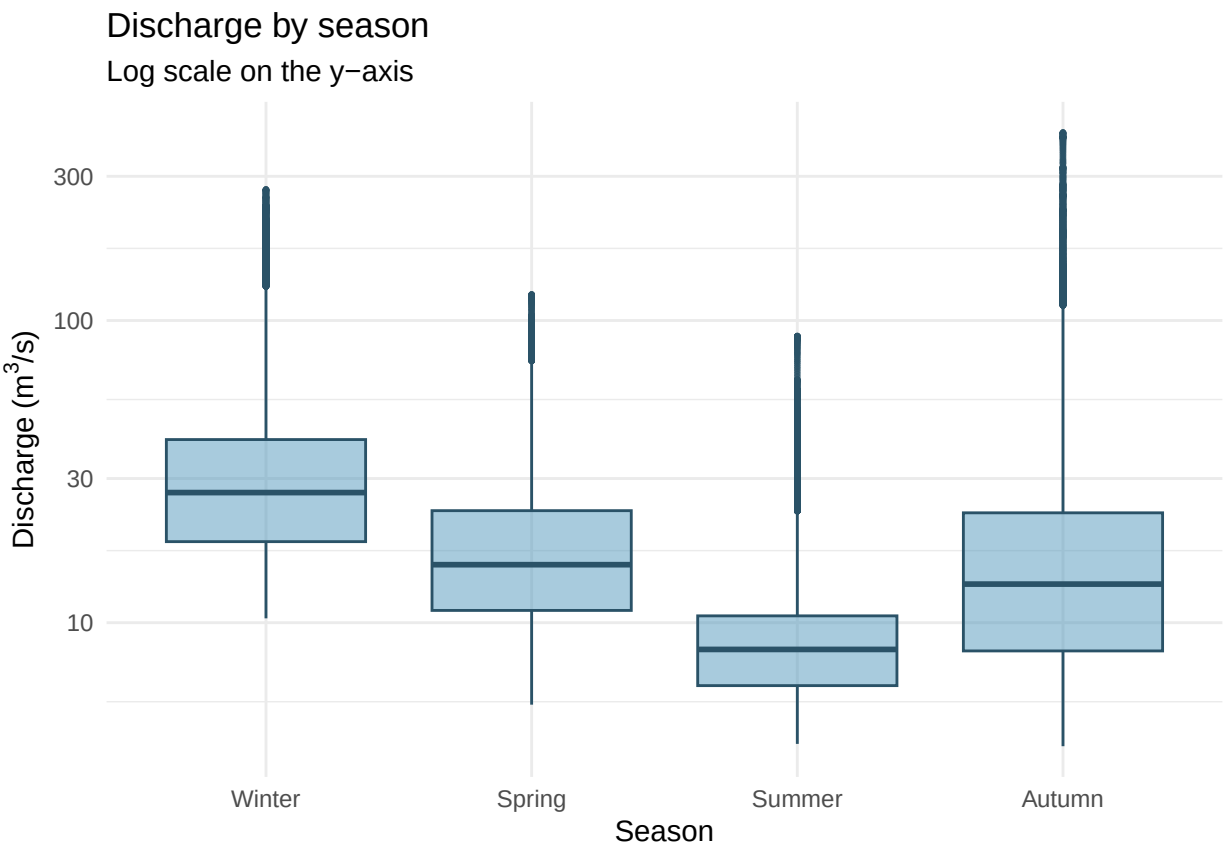



Table 5: Discharge by season ($m^3/s$)

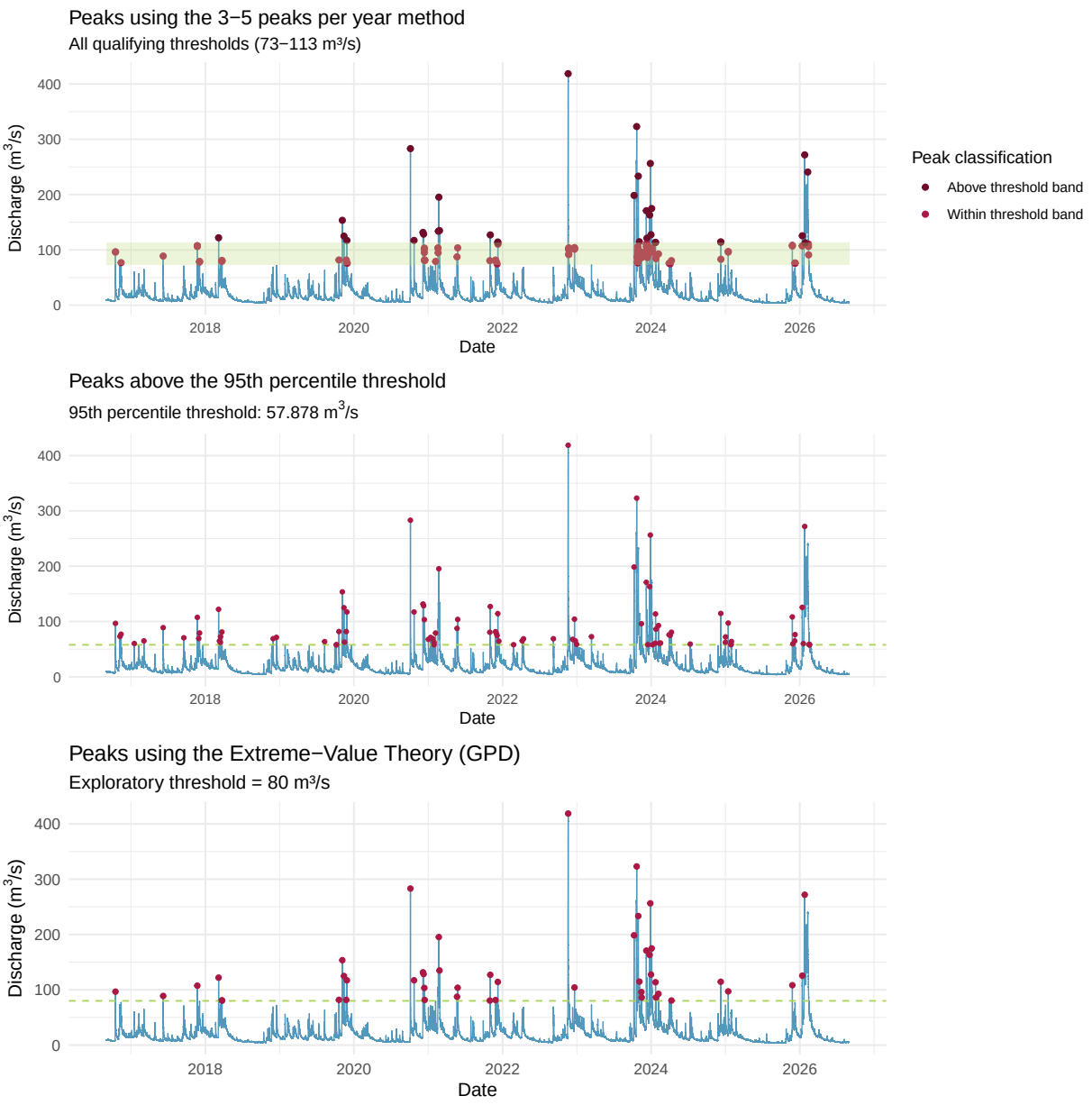
| Season | Mean | Median | 95th percentile | Maximum |
|---|---|---|---|---|
| Winter | 35.93 | 26.95 | 98.58 | 271.84 |
| Spring | 19.00 | 15.56 | 40.43 | 121.98 |
| Summer | 9.54 | 8.16 | 20.08 | 88.89 |
| Autumn | 22.49 | 13.42 | 69.48 | 418.68 |




## Threshold Selection

The Peaks-Over-Threshold (POT) uses a threshold to extracts the distinct peak flows for extreme events. The threshold is typically selected from three main methods:

1. 3-5 per year method
2. 95th Percentile method
3. Mean residual life threshold analysis




The figures above are a comparison of the thresholds selected by each method. Prior to independence analysis, a practical approach is to choose an initial threshold that is likely to be too low, i.e. ($u = 65$). This means when the distinct extreme events are selected, all the major peaks are being analysed, and subsequentially the threshold can be raised.